Table 8-3 **Wave properties for TE and TM modes in a rectangular waveguide with dimensions $a \times b$ that are filled with a dielectric material with constitutive parameters ε and μ. The TEM case—shown here for reference—pertains to plane-wave propagation in an unbounded medium.**

Rectangular Waveguides		Plane Wave
TE Modes	TM Modes	TEM Mode
$\widetilde{E}_x = \frac{j\omega\mu}{k_c^2}\left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)\ e^{-j\beta z}$	$\widetilde{E}_x = \frac{-j\beta}{k_c^2}\left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)\ e^{-j\beta z}$	$\widetilde{E}_x = E_{x0} e^{-j\beta z}$
$\widetilde{E}_y = \frac{-j\omega\mu}{k_c^2}\left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)\ e^{-j\beta z}$	$\widetilde{E}_y = \frac{-j\beta}{k_c^2}\left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)\ e^{-j\beta z}$	$\widetilde{E}_y = E_{y0} e^{-j\beta z}$
$\widetilde{E}_z = 0$	$\widetilde{E}_z = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)\ e^{-j\beta z}$	$\widetilde{E}_z = 0$
$\widetilde{H}_x = -\widetilde{E}_y / Z_{TE}$	$\widetilde{H}_x = -\widetilde{E}_y / Z_{TM}$	$\widetilde{H}_x = -\widetilde{E}_y / \eta$
$\widetilde{H}_y = \widetilde{E}_x / Z_{TE}$	$\widetilde{H}_y = \widetilde{E}_x / Z_{TM}$	$\widetilde{H}_y = \widetilde{E}_x / \eta$
$\widetilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)\ e^{-j\beta z}$	$\widetilde{H}_z = 0$	$\widetilde{H}_z = 0$
$Z_{TE} = \eta / \sqrt{1-(f_c/f)^2}$	$Z_{TM} = \eta \sqrt{1-(f_c/f)^2}$	$\eta = \sqrt{\mu/\varepsilon}$
Properties Common to TE and TM Modes		
$f_c = \frac{u_{p0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$		$f_c =$ not applicable
$\beta = k\sqrt{1-(f_c/f)^2}$		$k = \omega\sqrt{\mu\varepsilon}$
$u_p = \frac{\omega}{\beta} = u_{p0}/\sqrt{1-(f_c/f)^2}$		$u_{p0} = 1/\sqrt{\mu\varepsilon}$

Another difference between the TE and TM modes relates to the expression for the wave impedance. For TE,

$$Z_{TE} = \frac{\widetilde{E}_x}{\widetilde{H}_y} = -\frac{\widetilde{E}_y}{\widetilde{H}_x} = \frac{\eta}{\sqrt{1-(f_{mn}/f)^2}}. \qquad (8.111)$$

A summary of the expressions for the various wave attributes of TE and TM modes is given in **Table 8-3**. As a reference, corresponding expressions for the TEM mode on a coaxial transmission line are included as well.

Example 8-9: Cutoff Frequencies

For a hollow rectangular waveguide with dimensions $a = 3$ cm and $b = 2$ cm, determine the cutoff frequencies for all modes up to 20 GHz. Over what frequency range will the guide support the propagation of a single dominant mode?

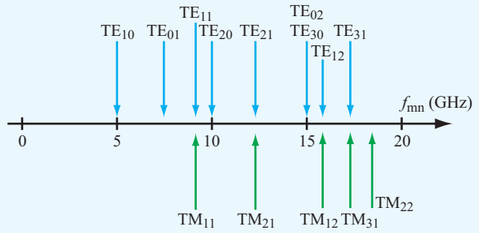


Figure 8-24 Cutoff frequencies for TE and TM modes in a hollow rectangular waveguide with $a = 3$ cm and $b = 2$ cm (Example 8-9).

Solution: A hollow guide has $\mu = \mu_0$ and $\varepsilon = \varepsilon_0$. Hence, $u_{p0} = 1/\sqrt{\mu_0\varepsilon_0} = c$. Application of Eq. (8.106) gives the cutoff frequencies shown in **Fig. 8-24**, which start at 5 GHz for the TE_{10} mode. To avoid all other modes, the frequency of operation should be restricted to the 5–7.5 GHz range.

表 8-3 尺寸为 a × b 的矩形波导中 TE 和 TM 模式的波特性，该波导填充有本构参数为 ε 和 μ 的介电材料。 TEM 案例（此处显示仅供参考）涉及无界介质中的平面波传播。

Rectangular Waveguides		Plane Wave
TE Modes	TM Modes	TEM Mode
$\widetilde{E}_x = \frac{j\omega\mu}{k_c^2}\left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)\ e^{-j\beta z}$	$\widetilde{E}_x = \frac{-j\beta}{k_c^2}\left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)\ e^{-j\beta z}$	$\widetilde{E}_x = E_{x0} e^{-j\beta z}$
$\widetilde{E}_y = \frac{-j\omega\mu}{k_c^2}\left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)\ e^{-j\beta z}$	$\widetilde{E}_y = \frac{-j\beta}{k_c^2}\left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)\ e^{-j\beta z}$	$\widetilde{E}_y = E_{y0} e^{-j\beta z}$
$\widetilde{E}_z = 0$	$\widetilde{E}_z = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)\ e^{-j\beta z}$	$\widetilde{E}_z = 0$
$\widetilde{H}_x = -\widetilde{E}_y / Z_{TE}$	$\widetilde{H}_x = -\widetilde{E}_y / Z_{TM}$	$\widetilde{H}_x = -\widetilde{E}_y / \eta$
$\widetilde{H}_y = \widetilde{E}_x / Z_{TE}$	$\widetilde{H}_y = \widetilde{E}_x / Z_{TM}$	$\widetilde{H}_y = \widetilde{E}_x / \eta$
$\widetilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)\ e^{-j\beta z}$	$\widetilde{H}_z = 0$	$\widetilde{H}_z = 0$
$Z_{TE} = \eta / \sqrt{1-(f_c/f)^2}$	$Z_{TM} = \eta \sqrt{1-(f_c/f)^2}$	$\eta = \sqrt{\mu/\varepsilon}$
Properties Common to TE and TM Modes		
$f_c = \frac{u_{p0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$		$f_c =$ not applicable
$\beta = k\sqrt{1-(f_c/f)^2}$		$k = \omega\sqrt{\mu\varepsilon}$
$u_p = \frac{\omega}{\beta} = u_{p0}/\sqrt{1-(f_c/f)^2}$		$u_{p0} = 1/\sqrt{\mu\varepsilon}$

TE 和 TM 模式之间的另一个区别与波阻抗的表达式有关。对于TE来说，

$$Z_{TE} = \frac{\widetilde{E}_x}{\widetilde{H}_y} = -\frac{\widetilde{E}_y}{\widetilde{H}_x} = \frac{\eta}{\sqrt{1-(f_{mn}/f)^2}}. \qquad (8.111)$$

表 8-3 总结了 TE 和 TM 模式的各种波属性的表达式。作为参考，还包括同轴传输线上 TEM 模式的相应表达式。

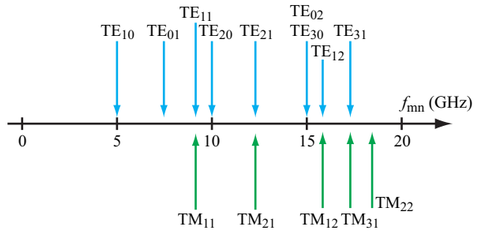


图 8-24 a = 3 cm 和 b = 2 cm 的空心矩形波导中 TE 和 TM 模式的截止频率（示例 8-9）。

例 8-9： 截止频率

对于尺寸 a = 3 cm 和 b = 2 cm 的空心矩形波导，确定高达 20 GHz 的所有模式的截止频率。波导将在什么频率范围内支持单一主模的传播？

解：空心导轨有$\mu = \mu_0$和$\varepsilon = \varepsilon_0$。因此，$u_{p0} = 1/\sqrt{\mu_0\varepsilon_0} = c$。方程的应用(8.106) 给出了如图 8-24 所示的截止频率，TE_{10} 模式的起始频率为 5 GHz。为了避免所有其他模式，工作频率应限制在 5–7.5 GHz 范围内。

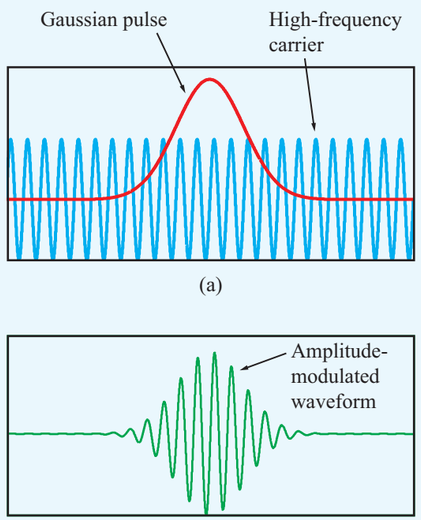


Figure 8-25 The amplitude-modulated high-frequency waveform in (b) is the product of the Gaussian-shaped pulse with the sinusoidal high-frequency carrier in (a).

8-10 Propagation Velocities

When a wave is used to carry a message through a medium or along a transmission line, information is encoded into the wave's amplitude, frequency, or phase. A simple example is shown in **Fig. 8-25**, where a high-frequency sinusoidal wave of frequency f is amplitude-modulated by a low-frequency Gaussian pulse. The waveform in **Fig. 8-25(b)** is the result of multiplying the Gaussian pulse shape in **Fig. 8-25(a)** by the carrier waveform.

By Fourier analysis, the waveform in **Fig. 8-25(b)** is equivalent to the superposition of a *group* of sinusoidal waves with specific amplitudes and frequencies. Exact equivalence may require a large (or infinite) number of frequency components, but in practice, it is often possible to represent the modulated waveform to a fairly high degree of fidelity with a wave group that extends over a relatively narrow bandwidth surrounding the high-frequency carrier f. The velocity that the envelope—or equivalently the wave group—travels through the medium is called the ***group velocity*** u_g. As such, u_g is the velocity of the energy carried by the wave group and of the information encoded in it. Depending on whether or not the propagation medium is dispersive, u_g may or may not be equal to the phase velocity u_p. In Section 2-1.1, we described a dispersive transmission line as one "on which the phase velocity is not a constant as a function of frequency," and as a result the shape of a pulse transmitted through it gets progressively distorted as it moves down the line. A rectangular waveguide constitutes a dispersive transmission line because the phase velocity of a TE or TM mode propagating through it is a strong function of frequency [per Eq. (8.108)], particularly at frequencies close to the cutoff frequency f_{mn}. As we see shortly, if $f \gg f_{mn}$, the TE and TM modes become approximately TEM in character—not only in terms of the directional arrangement of the electric and magnetic fields but also in terms of the frequency dependence of the phase velocity.

We now examine u_p and u_g in more detail. The phase velocity defined as the velocity of the sinusoidal pattern of the wave is given by

$$u_p = \frac{\omega}{\beta}, \tag{8.112}$$

while the group velocity u_g is given by

$$u_g = \frac{1}{d\beta/d\omega}. \tag{8.113}$$

Even though we will not derive Eq. (8.113) in this book, it is nevertheless important that we understand its properties for TE and TM modes in a metal waveguide. Using the expression for β given by Eq. (8.107),

$$u_g = \frac{1}{d\beta/d\omega} = u_{p0}\sqrt{1-(f_{mn}/f)^2}, \tag{8.114}$$

where—as before—u_{p0} is the phase velocity in an unbounded dielectric medium. In view of Eq. (8.108) for the phase velocity u_p,

$$u_p u_g = u_{p0}^2. \tag{8.115}$$

Above cutoff ($f > f_{mn}$), $u_p \geq u_{p0}$ and $u_g \leq u_{p0}$. As $f \to \infty$, or more precisely as $(f_{mn}/f) \to 0$, TE and TM modes approach the TEM case for which $u_p = u_g = u_{p0}$.

A useful graphical tool for describing the propagation properties of a medium or transmission line is the ***ω-β diagram***. In **Fig. 8-26**, the straight line starting at the origin represents the ω-β relationship for a TEM wave propagating in an unbounded medium (or on a TEM transmission line). The TEM line provides a reference to which the ω-β curves of the TE/TM modes can be compared. At a given location on the ω-β line or curve, the ratio of the value of ω to that of

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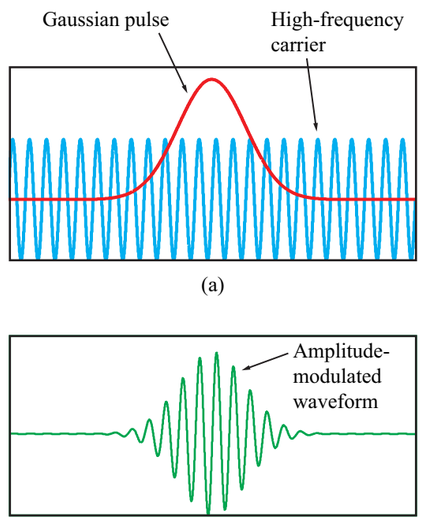


图 8-25 (b) 中的调幅高频波形是 (a) 中的高斯形脉冲与正弦高频载波的乘积。

8-10 传播速度

当波用于通过介质或沿着传输线携带消息时，信息被编码为波的幅度、频率或相位。一个简单的例子如图8-25所示，其中频率为f的高频正弦波由低频高斯脉冲进行幅度调制。图8-25(b)中的波形是将图8-25(a)中的高斯脉冲形状乘以载波波形的结果。

通过傅里叶分析，图8-25（b）中的波形相当于一组具有特定幅度和频率的正弦波的叠加。精确等效可能需要大量（或无限）的频率分量，但在实践中，通常可以使用在高频载波 f 周围相对较窄的带宽上延伸的波群来表示相当高保真度的调制波形。包络线（或等效的波群）穿过介质的速度称为群速度 u_g。因此，u_g 是波群携带的能量和信息的速度

编码在其中。根据传播介质是否色散，u_g 可能等于也可能不等于相速度 u_p。在第 2-1.1 节中，我们将色散传输线描述为“相速度不是频率函数的常数”，因此通过它传输的脉冲的形状随着沿线移动而逐渐失真。矩形波导构成色散传输线，因为通过它传播的 TE 或 TM 模式的相速度是频率的强函数 [根据方程 1]。(8.108)]，特别是在接近截止频率 f_{mn} 的频率下。正如我们很快看到的，如果 $f \gg f_{mn}$，TE 和 TM 模式在特性上变得近似 TEM——不仅在电场和磁场的方向排列方面，而且在相速度的频率依赖性方面。

我们现在更详细地研究 u_p 和 u_g。相速度定义为波的正弦模式的速度，由下式给出

$$u_p = \frac{\omega}{\beta}, \tag{8.112}$$

而群速度 u_g 则由下式给出

$$u_g = \frac{1}{d\beta/d\omega}. \tag{8.113}$$

即使我们不会推导出方程。（8.113）在本书中，我们了解金属波导中 TE 和 TM 模式的特性仍然很重要。使用等式给出的 β 表达式。(8.107),

$$u_g = \frac{1}{d\beta/d\omega} = u_{p0}\sqrt{1-(f_{mn}/f)^2}, \tag{8.114}$$

其中，与之前一样，u_{p0} 是无界电介质中的相速度。鉴于方程。(8.108) 对于相速度 u_p，

$$u_p u_g = u_{p0}^2. \tag{8.115}$$

高于截止值 ($f > f_{mn}$)、$u_p \geq u_{p0}$ 和 $u_g \leq u_{p0}$。当 $f \to \infty$ 时，或者更精确地作为 (f_{mn}/f) $\to 0$ 时，TE 和 TM 模式接近 TEM 情况，其中 $u_p = u_g = u_{p0}$。

ω-β 图是描述介质或传输线的传播特性的有用图形工具。在图8-26中，从原点开始的直线表示TEM波在无界介质（或TEM传输线上）传播的ω-β关系。 TEM 线提供了可以比较 TE/TM 模式的 ω-β 曲线的参考。在 ω-β 直线或曲线上的给定位置，ω 的值与

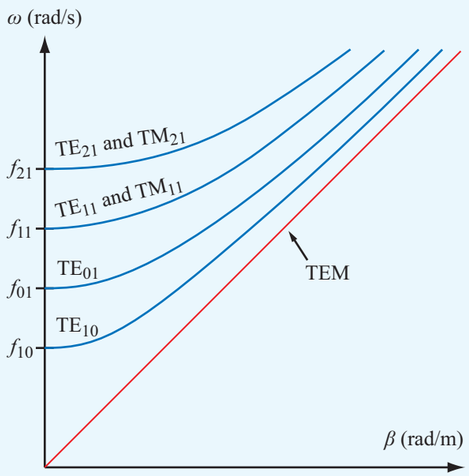


Figure 8-26 ω-β diagram for TE and TM modes in a hollow rectangular waveguide. The straight line pertains to propagation in an unbounded medium or on a TEM transmission line.

β defines $u_p = \omega/\beta$, whereas it is the slope $d\omega/d\beta$ of the curve at that point that defines the group velocity u_g. For the TEM line, the ratio and the slope have identical values (hence, $u_p = u_g$), and the line starts at $\omega = 0$. In contrast, the curve for each of the indicated TE/TM modes starts at a cutoff frequency specific to that mode below which the waveguide cannot support the propagation of a wave in that mode. At frequencies close to cutoff, u_p and u_g assume very different values; in fact, at cutoff $u_p = \infty$ and $u_g = 0$. On the other end of the frequency spectrum—at frequencies much higher than f_{mn}—the ω-β curves of the TE/TM modes approach the TEM line. We should note that for TE and TM modes, u_p may easily exceed the speed of light, but u_g will not. Since it is u_g that represents the actual transport of energy, Einstein's assertion that there is an upper bound on the speed of physical phenomena is not violated.

So far, we have described the fields in the guide, but we have yet to interpret them in terms of plane waves that zigzag along the guide through successive reflections. To do just that, consider the simple case of a TE_{10} mode. For $m = 1$ and $n = 0$, the only nonzero component of the electric field given by Eq. (8.110) is $\widetilde{E}_y$, so

$$\widetilde{E}_y = -j\frac{\omega\mu}{k_c^2}\left(\frac{\pi}{a}\right)H_0 \sin\left(\frac{\pi x}{a}\right)e^{-j\beta z}. \tag{8.116}$$

Using the identity $\sin\theta = (e^{j\theta} - e^{-j\theta})/2j$ for any argument θ, we obtain

$$\begin{aligned}\widetilde{E}_y &= \left(\frac{\omega\mu\pi H_0}{2k_c^2 a}\right)(e^{-j\pi x/a} - e^{j\pi x/a})e^{-j\beta z}\\ &= E_0'(e^{-j\beta(z+\pi x/\beta a)} - e^{-j\beta(z-\pi x/\beta a)})\\ &= E_0'(e^{-j\beta z'} - e^{-j\beta z''}),\end{aligned} \tag{8.117}$$

where we have consolidated the quantities multiplying the two exponential terms into the constant E_0'. The first exponential term represents a wave with propagation constant β traveling in the z' direction, where

$$z' = z + \frac{\pi x}{\beta a}, \tag{8.118a}$$

and the second term represents a wave traveling in the z'' direction with

$$z'' = z - \frac{\pi x}{\beta a}. \tag{8.118b}$$

From the diagram shown in **Fig. 8-27(a)**, it is evident that the z' direction is at an angle θ' relative to z, and the z'' direction is at an angle $\theta'' = -\theta'$. This means that the electric field $\widetilde{E}_y$ (and its associated magnetic field $\widetilde{\mathbf{H}}$) of the TE_{10} mode is composed of two TEM waves shown in **Fig. 8-27(b)** with both traveling in the $+z$ direction by zigzagging between the opposite walls of the waveguide. Along the zigzag directions (z' and z''), the phase velocity of the individual wave components is u_{p_0}, but the phase velocity of the combination of the two waves along z is u_p.

Example 8-10: Zigzag Angle

For the TE_{10} mode, express the zigzag angle θ' in terms of the ratio (f/f_{10}), and then evaluate it at $f = f_{10}$ and for $f \gg f_{10}$.

Solution: From **Fig. 8-27**,

$$\theta_{10}' = \tan^{-1}\left(\frac{\pi}{\beta_{10}a}\right),$$

where the subscript 10 has been added as a reminder that the expression applies to the TE_{10} mode specifically. For $m = 1$ and $n = 0$, Eq. (8.106) reduces to $f_{10} = u_{p_0}/2a$. After replacing β with the expression given by Eq. (8.107) and replacing a with $u_{p_0}/2f_{10}$, we obtain

$$\theta' = \tan^{-1}\left[\frac{1}{\sqrt{(f/f_{10})^2 - 1}}\right].$$

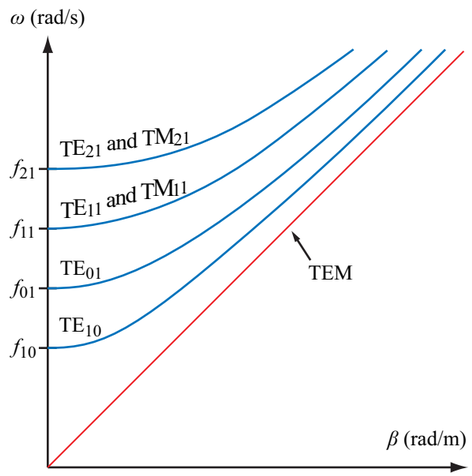


图 8-26 空心矩形波导中 TE 和 TM 模式的 ω-β 图。该直线适用于无界介质或 TEM 传输线上的传播。

β 定义了 $u_p = \omega/\beta$，而该点处曲线的斜率 $d\omega/d\beta$ 则定义了群速度 u_g。对于 TEM 线，比率和斜率具有相同的值（因此，$u_p = u_g$），并且线从 $\omega = 0$ 开始。相反，每个指示的 TE/TM 模式的曲线从特定于该模式的截止频率开始，低于该频率波导无法支持该模式下波的传播。在接近截止的频率处，u_p 和 u_g 呈现非常不同的值；事实上，在截止 $u_p = \infty$ 和 $u_g = 0$ 处。在频谱的另一端（在远高于 f_{mn} 的频率处），TE/TM 模式的 ω-β 曲线接近 TEM 线。我们应该注意到，对于 TE 和 TM 模式，u_p 可能很容易超过光速，但 u_g 不会。由于 u_g 代表了实际的能量传输，因此并不违反爱因斯坦关于物理现象速度存在上限的断言。

到目前为止，我们已经描述了波导中的场，但我们还没有用平面波来解释它们，平面波通过连续的反射沿着波导蜿蜒。为此，请考虑 TE_{10} 模式的简单情况。对于 $m = 1$ 和 $n = 0$，由方程式给出的电场的唯一非零分量。(8.110) 是 $\widetilde{E}_y$，所以

$$\widetilde{E}_y = -j\frac{\omega\mu}{k_c^2}\left(\frac{\pi}{a}\right)H_0 \sin\left(\frac{\pi x}{a}\right)e^{-j\beta z}. \tag{8.116}$$

使用恒等式 $\sin\theta = (e^{j\theta} - e^{-j\theta})/2j$ 对于任何参数 θ，我们得到

$$\begin{aligned}\widetilde{E}_y &= \left(\frac{\omega\mu\pi H_0}{2k_c^2 a}\right)(e^{-j\pi x/a} - e^{j\pi x/a})e^{-j\beta z}\\ &= E_0'(e^{-j\beta(z+\pi x/\beta a)} - e^{-j\beta(z-\pi x/\beta a)})\\ &= E_0'(e^{-j\beta z'} - e^{-j\beta z''}),\end{aligned} \tag{8.117}$$

我们将两个指数项相乘的量合并为常数 E_0'。第一个指数项表示传播常数为 β 的波，沿 z′ 方向传播，其中

$$z' = z + \frac{\pi x}{\beta a}, \tag{8.118a}$$

第二项表示沿 z″ 方向传播的波

$$z'' = z - \frac{\pi x}{\beta a}. \tag{8.118b}$$

从图8-27(a)所示的图中可以看出，z′方向相对于z成角度θ′，并且z″方向相对于z成角度$\theta'' = -\theta'$。这意味着 TE_{10} 模式的电场 $\widetilde{E}_y$（及其相关磁场 $\widetilde{H}$）由图 8-27(b) 中所示的两个 TEM 波组成，两个 TEM 波均在波导的相对壁之间锯齿形沿 +z 方向传播。沿之字形方向（z′ 和 z″），各个波分量的相速度为 u_{p_0}，但沿 z 的两个波组合的相速度为 u_p。

示例 8-10：Z 字形角度

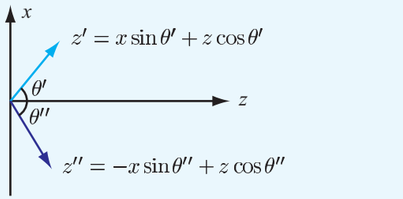
对于 TE_{10} 模式，用比率 (f/f_{10}) 表示锯齿角 θ′，然后在 $f = f_{10}$ 和 $f \gg f_{10}$ 处对其进行评估。

解：由图8-27可知，

$$\theta_{10}' = \tan^{-1}\left(\frac{\pi}{\beta_{10}a}\right),$$

其中添加下标 10 是为了提醒该表达式特别适用于 TE_{10} 模式。对于 $m = 1$ 和 $n = 0$，等式：(8.106) 简化为 $f_{10} = u_{p_0}/2a$。将 β 替换为等式给出的表达式后：(8.107) 并将 a 替换为 $u_{p_0}/2f_{10}$，我们得到

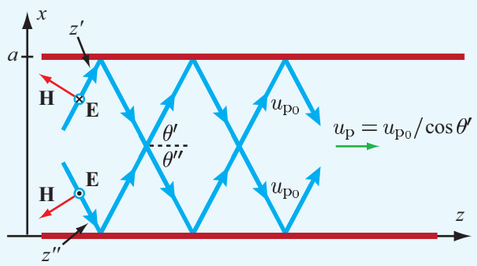
$$\theta' = \tan^{-1}\left[\frac{1}{\sqrt{(f/f_{10})^2 - 1}}\right].$$



From Eq. (8.118a), $z' = \frac{\pi x}{\beta a} + z$.
Hence, $\theta' = \tan^{-1}(\pi/\beta a)$.

From Eq. (8.118b), $z'' = -\frac{\pi x}{\beta a} + z$.
Hence, $\theta'' = -\tan^{-1}(\pi/\beta a)$.

(a) z' and z'' propagation directions



(b) TEM waves

Figure 8-27 The TE_{10} mode can be constructed as the sum of two TEM waves.

At $f = f_{10}$, $\theta' = 90°$, which means that the wave bounces back and forth at normal incidence between the two side walls of the waveguide, making no progress in the z direction. At the other end of the frequency spectrum, when $f \gg f_{10}$, θ' approaches 0 and the wave becomes TEM-like as it travels straight down the guide.

Concept Question 8-12: For TE waves, the dominant mode is TE_{10}, but for TM the dominant mode is TM_{11}. Why is it not TM_{10}?

Concept Question 8-13: Why is it acceptable for u_p to exceed the speed of light c, but not so for u_g?

Exercise 8-13: What do the wave impedances for TE and TM look like as f approaches f_{mn}?

Answer: At $f = f_{mn}$, Z_{TE} looks like an open circuit, and Z_{TM} looks like a short circuit.

Exercise 8-14: What are the values for (a) u_p, (b) u_g, and (c) the zigzag angle θ' at $f = 2f_{10}$ for a TE_{10} mode in a hollow waveguide?

Answer: (a) $u_p = 1.15c$, (b) $u_g = 0.87c$, (c) $\theta' = 30°$.

8-11 Cavity Resonators

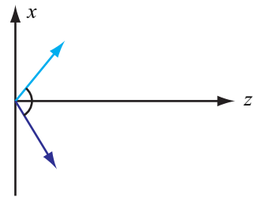
A rectangular waveguide has metal walls on four sides. When the two remaining sides are terminated with conducting walls, the waveguide becomes a cavity. By designing cavities to *resonate* at specific frequencies, they can be used as circuit elements in ***microwave oscillators***, ***amplifiers***, and ***bandpass filters***.

The rectangular cavity shown in **Fig. 8-28(a)** with dimensions $(a \times b \times d)$ is connected to two coaxial cables that feed and extract signals into and from the cavity via input and output probes. As a bandpass filter, the function of a resonant cavity is to block all spectral components of the input signal except for those with frequencies that fall within a narrow band surrounding a specific center frequency f_0, which is the cavity's ***resonant frequency***. Comparison of the spectrum in **Fig. 8-28(b)**, which describes the range of frequencies that might be contained in a typical input signal, with the narrow output spectrum in **Fig. 8-28(c)** demonstrates the filtering action imparted by the cavity.

In a rectangular waveguide, the fields constitute standing waves along the x and y directions and a propagating wave along $\hat{\mathbf{z}}$. The terms TE and TM were defined relative to the propagation direction; TE meant that $\mathbf{E}$ was entirely transverse to $\hat{\mathbf{z}}$, and TM meant that $\mathbf{H}$ had no component along $\hat{\mathbf{z}}$. In a cavity, there is no unique propagation direction, as *no* fields propagate. Instead, standing waves exist along all three directions. Hence, the terms TE and TM need to be modified by defining the fields relative to one of the three rectangular axes. For the sake of consistency, *we will continue to define the transverse direction to be any direction contained in the plane whose normal is* $\hat{\mathbf{z}}$.

The TE mode in the rectangular waveguide consists of a single propagating wave whose $\widetilde{H}_z$ component is given by Eq. (8.110e) as

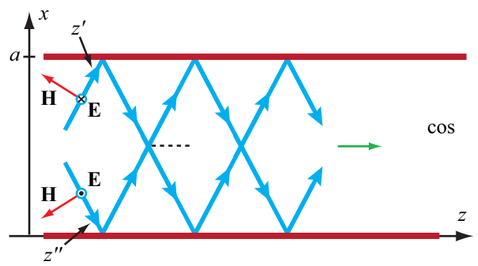
$$\widetilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \qquad (8.119)$$



From Eq. (8.118a), — .
Hence, .

From Eq. (8.118b), — .
Hence, .

(a) z' and z'' propagation directions



(b) TEM waves

图 8-27 TE_{10} 模式可以构造为两个 TEM 波的总和。

在 f = f_{10}、θ′ = 90° 处，这意味着波在波导的两个侧壁之间以法向入射来回反射，在 z 方向上没有前进。在频谱的另一端，当 f ≫ f_{10}、θ′ 接近 0 时，波在沿着波导直接传播时变得类似 TEM。

概念问题8-12：对于TE波，主模是TE_{10}，但对于TM波，主模是TM_{11}。为什么不是TM_{10}?

概念问题 8-13：为什么 u_p 超过光速 c 可以接受，而 u_g 则不能?

练习8-13：当f接近f_{mn}时，TE和TM的波阻抗是什么样的?

答案：在 f = f_{mn} 处，Z_{TE} 看起来像开路，而 Z_{TM} 看起来像短路。

练习 8-14：空心波导中 TE_{10} 模式的 (a) u_p、(b) u_g 和 (c) f = 2 f_{10} 处的锯齿角 θ′ 的值是多少?

答案：(a) $u_p = 1.15c$，(b) $u_g = 0.87c$，(c) $\theta' = 30°$。

8-11 空腔谐振器

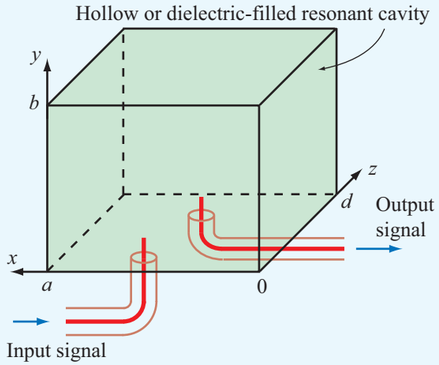
矩形波导的四个侧面都有金属壁。当其余两个侧面以导电壁终止时，波导就变成了空腔。通过设计在特定频率下谐振的腔体，它们可以用作微波振荡器、放大器和带通滤波器中的电路元件。

矩形腔体如图8-28(a)所示，尺寸为(a×乙×d) 连接到两根同轴电缆，通过输入和输出探头将信号馈入腔体或从腔体提取信号。作为带通滤波器，谐振腔的功能是阻挡输入信号的所有频谱分量，除了那些频率落在围绕特定中心频率 f 的窄带内的频谱分量。$_0$，这是腔体的谐振频率。图 8-28(b) 中的频谱描述了典型输入信号中可能包含的频率范围，与图 8-28(c) 中的窄输出频谱的比较表明了腔体所产生的滤波作用。

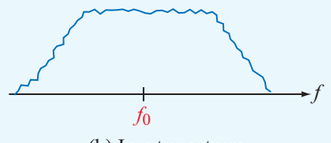
在矩形波导中，场沿 x 和 y 方向构成驻波，并沿 ẑ 构成传播波。术语 TE 和 TM 是相对于传播方向定义的； TE 表示 E 完全横向于 ẑ，TM 表示 H 没有沿 ẑ 的分量。在空腔中，没有唯一的传播方向，因为没有场传播。相反，驻波沿着所有三个方向都存在。因此，需要通过定义相对于三个矩形轴之一的场来修改术语 TE 和 TM。为了保持一致性，我们将继续将横向方向定义为法线为 ẑ 的平面中包含的任何方向。

矩形波导中的 TE 模式由单个传播波组成，其 $\widetilde{H}_z$ 分量由方程 1 给出。 (8.110e) 为

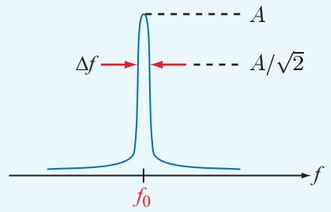
$$\widetilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \qquad (8.119)$$



(a) Resonant cavity



(b) Input spectrum



(c) Output spectrum

Figure 8-28 A resonant cavity supports a very narrow bandwidth around its resonant frequency f_0.

where the phase factor $e^{-j\beta z}$ signifies propagation along $+\hat{\mathbf{z}}$. Because the cavity has conducting walls at both $z = 0$ and $z = d$, it will contain two such waves: one with amplitude H_0 traveling along $+\hat{\mathbf{z}}$ and another with amplitude H_0^- traveling along $-\hat{\mathbf{z}}$. Hence,

$$\widetilde{H}_z = (H_0 e^{-j\beta z} + H_0^- e^{j\beta z}) \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right). \quad (8.120)$$

Boundary conditions require the normal component of $\widetilde{\mathbf{H}}$ to be zero at a conducting boundary. Consequently, $\widetilde{H}_z$ must be zero at $z = 0$ and $z = d$. To satisfy these conditions, it is necessary that $H_0^- = -H_0$ and $\beta d = p\pi$, with $p = 1, 2, 3, \ldots$. In this case, Eq. (8.120) becomes

$$\widetilde{H}_z = -2jH_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) \sin\left(\frac{p\pi z}{d}\right). \quad (8.121)$$

Given that $\widetilde{E}_z = 0$ for the TE modes, all of the other components of $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ can be derived readily through the application of the relationships given by Eq. (8.89). A similar procedure also can be used to characterize cavity modes for the TM case.

8-11.1 Resonant Frequency

The consequence of the quantization condition imposed on β, namely $\beta = p\pi/d$ with p assuming only integer values, is that for any specific set of integer values of (m, n, p) the wave inside the cavity can exist at only a single ***resonant frequency***, f_{mnp}, whose value has to satisfy Eq. (8.105). The resulting expression for f_{mnp} is

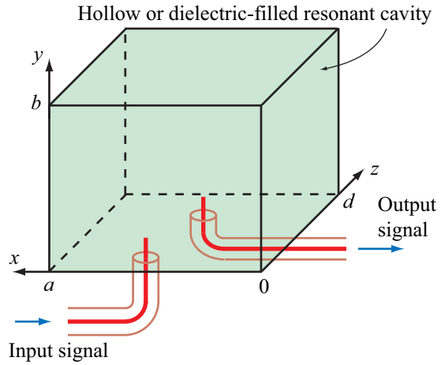
$$f_{mnp} = \frac{u_{p_0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}. \quad (8.122)$$

For TE, the indices m and n start at 0, but p starts at 1. The exact opposite applies to TM. By way of an example, the resonant frequency for a TE_{101} mode in a hollow cavity with dimensions $a = 2$ cm, $b = 3$ cm, and $d = 4$ cm is $f_{101} = 8.38$ GHz.

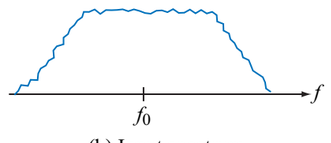
8-11.2 Quality Factor

In the ideal case, if a group of frequencies is introduced into the cavity to excite a certain TE or TM mode, only the frequency component at exactly f_{mnp} of that mode will survive, and all others will attenuate. If a probe is used to couple a sample of the resonant wave out of the cavity, the output signal will be a monochromatic sinusoidal wave at f_{mnp}. In practice, the cavity exhibits a frequency response similar to that shown in **Fig. 8-28(c)**, which is very narrow, but it is not a perfect spike. The bandwidth Δf of the cavity is defined as the frequency range between the two frequencies (on either side of f_{mnp}) at which the amplitude is $1/\sqrt{2}$ of the maximum amplitude (at f_{mnp}). The ***normalized bandwidth***, defined as $\Delta f/f_{mnp}$, is approximately equal to the reciprocal of the ***quality factor*** Q of the cavity:

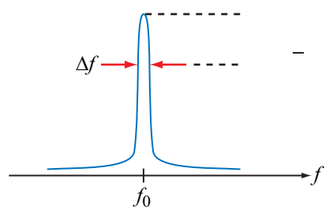
$$Q \approx \frac{f_{mnp}}{\Delta f}. \quad (8.123)$$



(a) Resonant cavity



(b) Input spectrum



(c) Output spectrum

图 8-28 谐振腔在其谐振频率 f_0 附近支持非常窄的带宽。

导电边界为零。因此，在 z = 0 和 z = d 处，$\widetilde{H}_z$ 必须为零。为了满足这些条件，需要 $H_0^- = -H_0$ 和 β d = pπ，其中 p = 1, 2, 3, … 。在这种情况下，等式。 (8.120) 变为

$$\widetilde{H}_z = -2jH_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) \sin\left(\frac{p\pi z}{d}\right). \quad (8.121)$$

给定 TE 模式的 $\widetilde{E}_z = 0$，$\widetilde{E}$ 和 $\widetilde{H}$ 的所有其他分量可以通过应用方程 1 给出的关系轻松导出。 （8.89）。类似的过程也可用于表征 TM 情况的腔模式。

8-11.1 谐振频率

对 β 施加量化条件（即 β = pπ/d，其中 p 仅假设整数值）的结果是，对于 (m, n, p) 的任何特定整数值集，腔内的波只能以单个谐振频率 f_{mnp} 存在，其值必须满足方程 1。（8.105）。 f_{mnp} 的结果表达式为

$$f_{mnp} = \frac{u_{p_0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}. \quad (8.122)$$

对于 TE，索引 m 和 n 从 0 开始，但 p 从 1 开始。TM 的情况正好相反。举例来说，尺寸为a=2cm、b=3cm和d=4 cm的空腔中TE_{101}模式的谐振频率为f_{101} =8.38GHz。

8-11.2 品质因数

在理想情况下，如果将一组频率引入腔体以激发某种 TE 或 TM 模式，则只有该模式的 f_{mnp} 处的频率分量会保留下来，所有其他频率分量都会衰减。如果使用探头将谐振波样本耦合到腔外，则输出信号将是 f_{mnp} 处的单色正弦波。实际上，空腔表现出类似于图8-28(c)所示的频率响应，该频率响应非常窄，但它并不是完美的尖峰。腔体的带宽 Δ f 定义为两个频率（f_{mnp} 两侧）之间的频率范围，在此范围内，幅度为最大幅度（在 f_{mnp} 处）的 $1/\sqrt{2}$。归一化带宽定义为 Δ f / f_{mnp}，约等于腔体品质因数 Q 的倒数：

$$Q \approx \frac{f_{mnp}}{\Delta f}. \quad (8.123)$$